

Qixin Li

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Education

University of Alberta, Ph.D. Student in Rehabilitation Science Sept 2026 – Present
Faculty of Rehabilitation Medicine

Beijing Normal University, Master of Arts in Theoretical Linguistics Sept 2023 – Jun 2026
Institute of Theoretical Linguistics

- **GPA:** 3.9/4.0
- **Selected Coursework:** Phonetics and Phonology, Phonetics and Artificial Intelligence, Cognitive-Functional Linguistics, Programming with Python, Early Language Development.

Beijing Normal University, Bachelor of Arts in Chinese Language and Literature Sept 2019 – Jun 2023
School of Chinese Language and Literature

- **Average Score:** 90.12/100
- **Selected Coursework:** Modern Chinese Phonetics, Modern Chinese Syntax, Ancient Chinese Language, Chinese Dialectology, Cognitive Linguistics.

Research Interests

I am a postgraduate researcher specializing in experimental phonetics, with a long-term interest in understanding how language is co-evolving with the powerful forces of AI adoption and linguistic standardization. My specific interests include:

- **Human-Computer Interaction:** Acoustic-prosodic entrainment in human-AI dialogues to improve collaboration.
- **Sociophonetics:** Gender bias in tone perception and language prestige in conversational adaptation.
- **Phonetics:** Prosody and its interfaces with pragmatics and syntax.

Publications

- Li, Q., Lu, H., Wang, G. (2025) *When AI Speaks, Do We Follow? Phonetic Entrainment in Human-AI Dialogues*. NCMMSC 2025, vol 2662. Springer, Singapore.

Analyzed 18-hour (40 pairs of) human-AI dialogues and revealed phonetic entrainment varied by task—with open-ended chat favoring immediate synchrony (intensity & speech rate) and collaborative planning eliciting stronger convergence (f0 & wav2vec2)—but showed no correlation with users' explicit attitudes.

- Li, Q. (2025). *Phonetic Entrainment in Short-Term Dialogues: A Case Study of Standard and Changsha Mandarin*. PCC.

Modeled phonetic convergence in 24 dyads using time-series correlation, supporting that language prestige and cognitive salience produced asymmetric, level-dependent convergence: prestige drove sentence-level convergence from dialect to standard, while salience drove a reversed word-level pattern. Sentence-level convergence was also identified as the optimal metric for individual differences, significantly correlating with social approval needs.

- Li, Q. Wang, G. (2024). *Focus and Gender Affect Sentence Type Perception: Observations on Mandarin Sentence-Final Particle Ba*. ISCSLP. doi: 10.1109/ISCSLP63861.2024.10800709.

Conducted acoustic analysis and dual perception tasks, finding through LME modeling that narrow focus increased question perception, and gender-ambiguous voices were most frequently perceived as questions.

- Li, Q. Wang, G. (2023). *Gender Differences in Mood and Modality Identification of Mandarin Sentences with Sentence-final Modal Particle Ba*. NCMMSC.

Used a forced-choice perception task with adults and children on synthesized stimuli and found that adults outperformed children; female voices were perceived as stronger in modality, and female listeners responded faster and more consistently.

- **Li, Q. Wang, G. (2023). *The Influence of Prosodic Features of Sentence-final Modal Particles Ba in Beijing Dialect on Mood and Modality Recognition*. PCC.**

Performed acoustic analysis (Praat) and perception experiment (E-Prime) on the particle ba and found that shorter duration marked declarative sentences, while longer duration and lower pitch reduced epistemic certainty.

- **He, Y., Li, Q., Wang, Y., & Wang, G. (2026). *Speech Rate Convergence in Human–AI Dialogue Across Multiple Emotional Scenarios*. IALP 2026, [Accepted]**

Examined speech rate convergence in 114 Mandarin human–AI dialogues involving 38 participants across neutral, argument, and consolation scenarios, finding greater overall convergence in argument than in consolation interactions.

- **Li, Q. (Unpublished Manuscript). *Interaction of Boundary Tone and Epistemic Stance in Mandarin Ba-Sentences*.**

Analyzed synthesized pitch continua and production data under stance manipulation and found that context outweighed pitch cues, and listeners distinguished binary confidence levels rather than gradient stance.

Work Experience

Speech Data Researcher

2noise Technology (ChatTTS Project)

Mar 2025 – Jun 2025

- Curated and developed large-scale, high-quality speech datasets for training and evaluating the ChatTTS text-to-speech engine, focusing on human-LLM interactions.

AI Speech Data Analyst

ByteDance Technology Co., Ltd. (Data-Audio Team)

Sep 2024 – Jan 2025

- Improved emotional expressiveness in LLM-based dialogue systems (e.g., *Doubao*) through specialized data development and evaluation.

Linguistic Data Specialist

DeepLang AI (Lingowhale Project)

Jul 2024 – Sep 2024

- Refined annotation guidelines for minimal information units to ensure inter-annotator consistency.
- Trained annotation teams and supervised final dataset construction.

Teaching Assistant

Department of Theoretical Linguistics, School of Chinese Language and Literature, BNU

Sep 2023 – Sep 2025

- Led tutorials and graded all coursework for two core courses: *Introduction to Phonetics* (undergraduate) and *Phonetics and Phonology* (graduate).
- Provided student mentorship and technical support for acoustic and statistic analysis.

Dialect Research Intern

Beijing Normal University (Dialect Survey Project)

Jun 2022 – Aug 2022

- Conducted fieldwork on the Renqiu dialect (Ji-Lu Mandarin, Cang-Huang subgroup).
- Authored a report on phonological, lexical, and grammatical features.

Projects

- **BNU National Language Teaching Reform Project: *PBL Curriculum Reform in Phonetics*** (2024.06–2025.06)
Designed case-based phonetics teaching modules, covering speech therapy, audiobook production, and AI speech technology in a project-based learning framework.
- **BNU Digital Curriculum Development Project: *Phonetics and Artificial Intelligence*** (2024.05–2025.03)
Lead author of the *Introduction to Python Programming* chapter, covering text segmentation, word frequency analysis, web scraping, speech feature extraction, TTS, and ASR with Python.

Honors & Awards

Best Oral Presentation Performer

2025

VUB International Winter Program, Vrije Universiteit Brussel, Belgium

Academic Scholarship(1st*1, 2nd*2, 3rd*1)

2024, 2022, 2021, 2020

Beijing Normal University

Skills

- **Languages:** Mandarin (native), English (IELTS 8.0), French/Spanish (beginner)
- **Programming:** Python
- **Experimental Software:** E-prime 3.0
- **Data Analysis:** Familiar with statistical analysis and data visualization using SPSS or R
- **Linguistic Tools:** Proficient in acoustic analysis and linguistic corpus annotation using Praat